

Real-Time Earthquake Monitoring & Analysis System

Mikołaj Malec 298828

Project Deliverables

1. Description of Project Outcomes

The primary deliverable of this project is a robust, scalable Real-Time Earthquake Monitoring System. From the user perspective, this system transforms high-volume, noisy seismic data from multiple global sources into a unified, reliable, and visually actionable dashboard.

The system is delivered via a web-based dashboard (Streamlit) featuring four key interactive widgets:

- Real-Time Event Feed: An auto-updating list of the most recent global events.
- Seismic Activity Map: A layered geospatial view of ML-generated historical clusters.
- Regional/Cluster Metrics: Charts showing the evolution of seismic intensity across fixed geographic regions and ML-generated historical clusters.

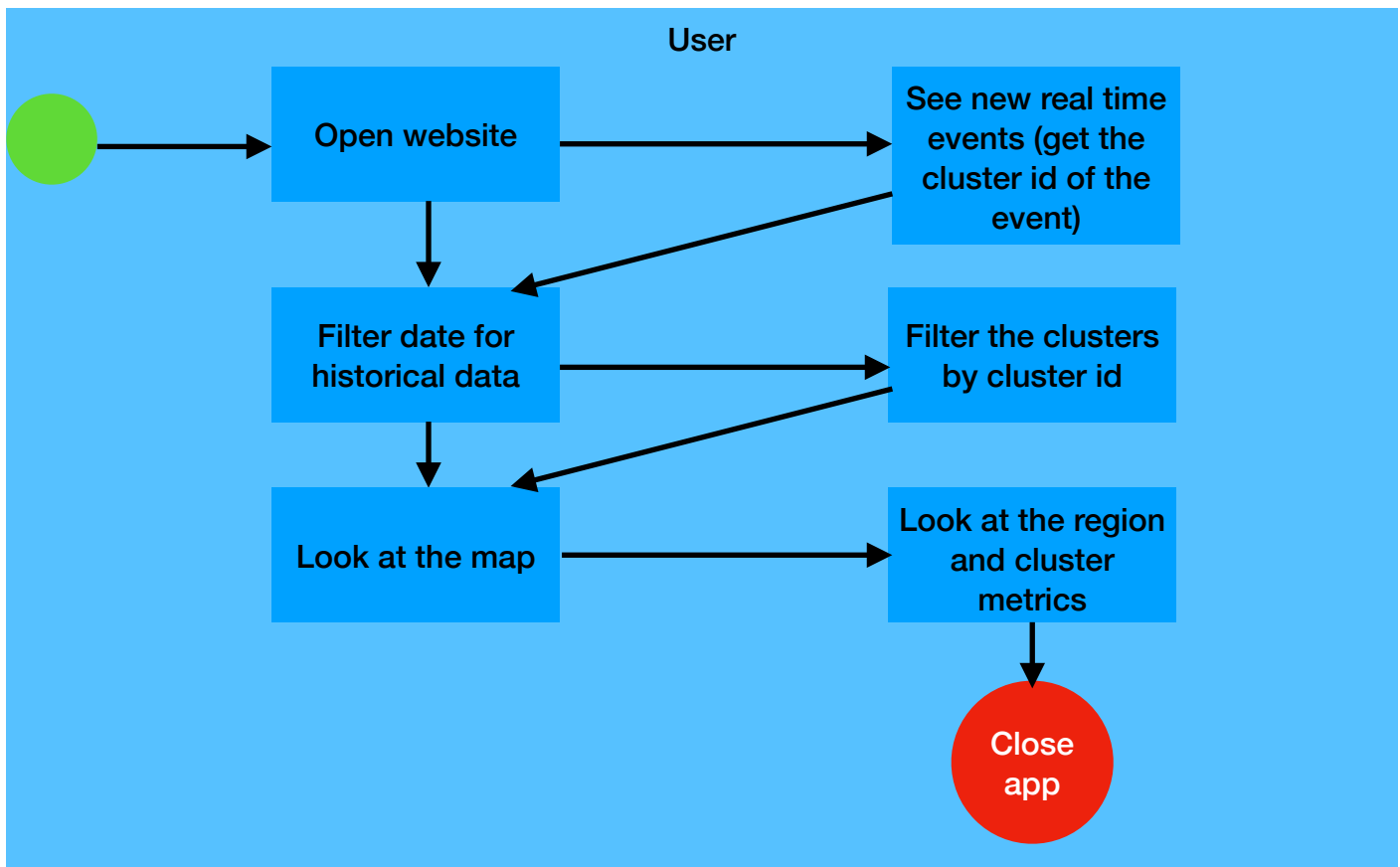
2. Anticipated Benefits

The system provides tailored benefits for its three primary user groups: Analysts, Seismologists, and Emergency Operations Officers.

- For the Emergency Operations Officer
 - Global Coverage: Access to a merged dataset ensures that events missed by one provider (e.g., USGS) might be caught by another (e.g., Terraquake), maximising safety coverage.
- For the Analyst
 - Data-Driven Reporting: The system automates the aggregation of daily statistics for regions, allowing analysts to get fast and refined data.
- For the Seismologist
 - Pattern Recognition: The K-Means clustering algorithm reveals non-obvious geographical relationships and fault lines that traditional static maps might obscure.

3. Business Process Diagram (Core Use Case)

The following diagram outlines the core workflow for a user engaging with the system to assess a seismic analysis.



Data Sets

1. Data Sources Overview

The system aggregates seismic data from two primary public APIs to ensure global coverage and reliability. To support the analytical modules, it also utilises a static reference dataset for geographical definitions.

Source Name	Type	Base Volume	Update Rate	Role
USGS Earthquake Hazards API	GeoJSON (REST)	~129,000 events (Historical) ~300 events/day (Live)	Near Real-Time (Seconds to Minutes)	Primary Authority. Provides the bulk of the high-confidence scientific data.
TerraQuake API	JSON (REST)	~11,000 events (Historical) ~50 events/day (Live)	Near Real-Time (~2-10s latency)	Supplementary. Used to validate USGS data and fill potential coverage gaps.
Region Definitions	CSV (Static)	~16 Records	Static	Reference. Defines fixed geographical boundaries (e.g., "Circum-Pacific") for reporting.

2. Conceptual Data Model

The system follows a Medallion Architecture, organising data into three progressively refined layers. This ensures a clear separation between raw ingestion, cleaned "truth," and business-ready aggregations.

- **Layer 1: Raw Data (Bronze Layer)**
 - Purpose: Acts as the immutable historical archive.
 - Storage: HDFS (Parquet), partitioned by date (dt).
 - Content: Stores the exact JSON/GeoJSON responses received from the APIs.
 - Validation: Ingestion pipelines (Apache NiFi) perform a preliminary check to ensure every record contains a valid creation_time before writing to storage.
- **Layer 2: Clean & Unified Data (Silver Layer)**
 - Purpose: Serves as the "Single Source of Truth" for the system.
 - Transformation:
 - Unification: Maps divergent schemas from USGS and Terraquake into a strict common schema (Standardising fields like time_utc, magnitude, depth_km, and coordinates).
 - Deduplication: Uses a Fuzzy Matching Strategy to merge records. Events from different providers are considered identical if they fall within the same 1-minute time bucket and ~11km geographic bucket. In conflicts, USGS data is prioritised.
 - Data Quality: The resulting dataset contains 140,008 unique events with exceptional completeness (0 null values in key columns like time, latitude, longitude, and depth).
- **Layer 3: Aggregated Data (Gold Layer)**
 - Purpose: Optimised tables ready for the Presentation Layer (Streamlit).
 - Structure: The data is aggregated into two distinct analytical views:
 - Regional Analysis View (Static): Aggregates daily statistics (Count, Max Magnitude, Avg Depth) based on fixed geographical boundaries (e.g., "Eurasia").
 - Cluster Analysis View (Dynamic): Aggregates data based on ML-generated clusters. Unlike fixed regions, these "hotspots" can move or change size based on the K-Means algorithm's daily retraining.

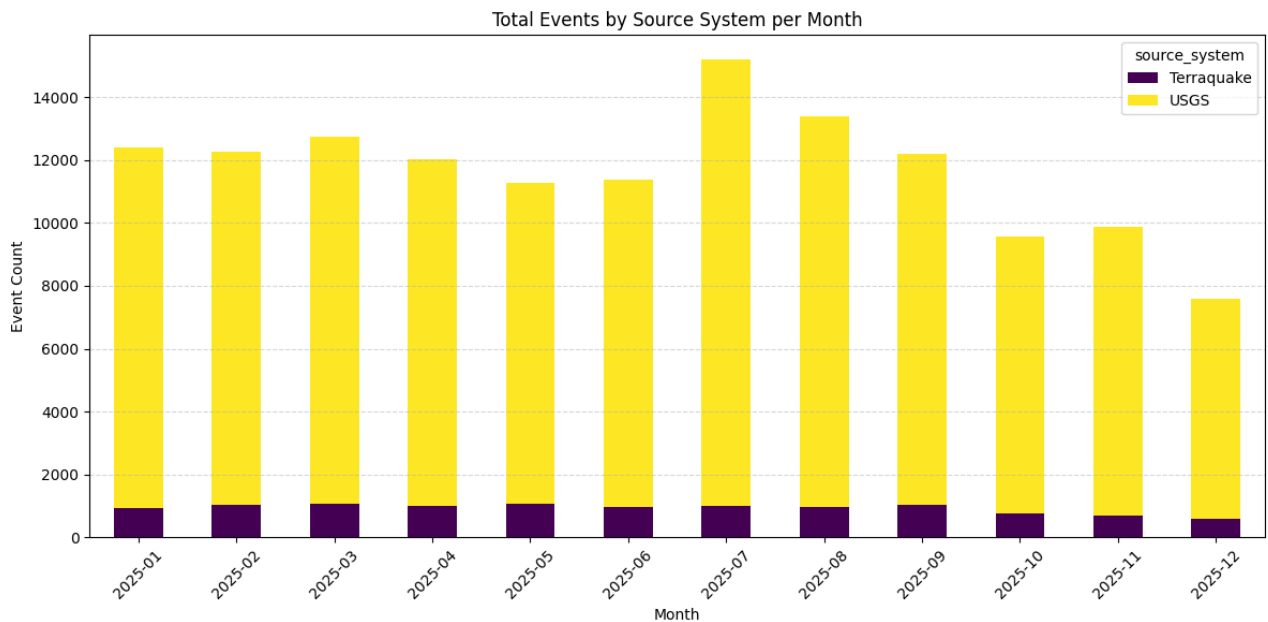
Table Name	Layer	Description	Key Attributes
earthquake_clean	Silver	The unified list of individual seismic events.	event_id, time_utc, magnitude, coordinates, source_system
fact_region_daily_stats	Gold	Daily summaries for fixed regions.	region_id, date, total_events, max_magnitude, avg_depth
fact_cluster_daily_stats	Gold	Daily summaries for ML-detected clusters.	cluster_id, date, density, centroid_lat, centroid_lon
dim_cluster_metadata	Gold	Defines the current active clusters (output of ML model).	cluster_id, creation_date, radius_km, wcss_score

Exploratory Data Analysis (EDA)

1. Dataset Overview

The EDA was performed on the Silver Layer (Clean Dataset), which serves as the "Single Source of Truth" for the system. This dataset represents the successful unification and deduplication of 140,008 raw records from the USGS and Terraquake APIs.

- Total Volume: 140,008 Unique Seismic Events.
- Source Contribution:
 - USGS: 92.1% (Primary Authority).
 - Terraquake: 7.9% (Supplementary Source).
- Completeness: The dataset exhibits exceptional quality with 0 null values in critical columns (time_utc, latitude, longitude, depth_km), validating the robustness of the ingestion pipeline.

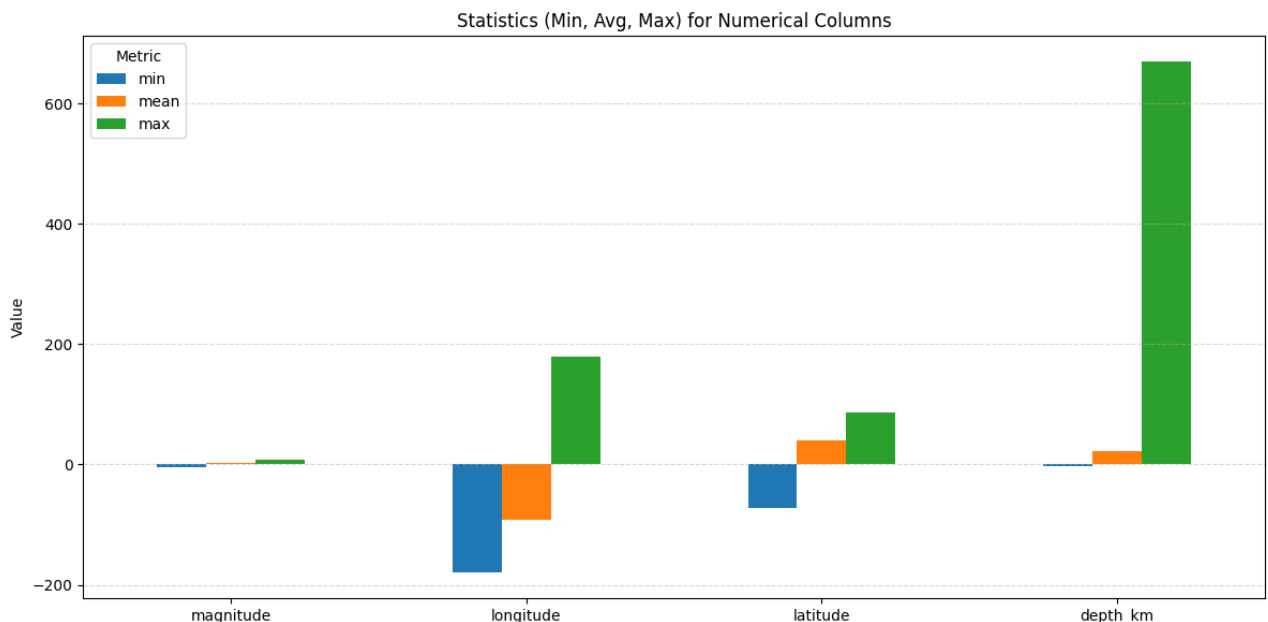


2. Statistical Distributions & Visualizations

Magnitude Distribution

The magnitude data reveals a classic power-law distribution consistent with the Gutenberg-Richter law, where minor tremors exponentially outnumber major earthquakes.

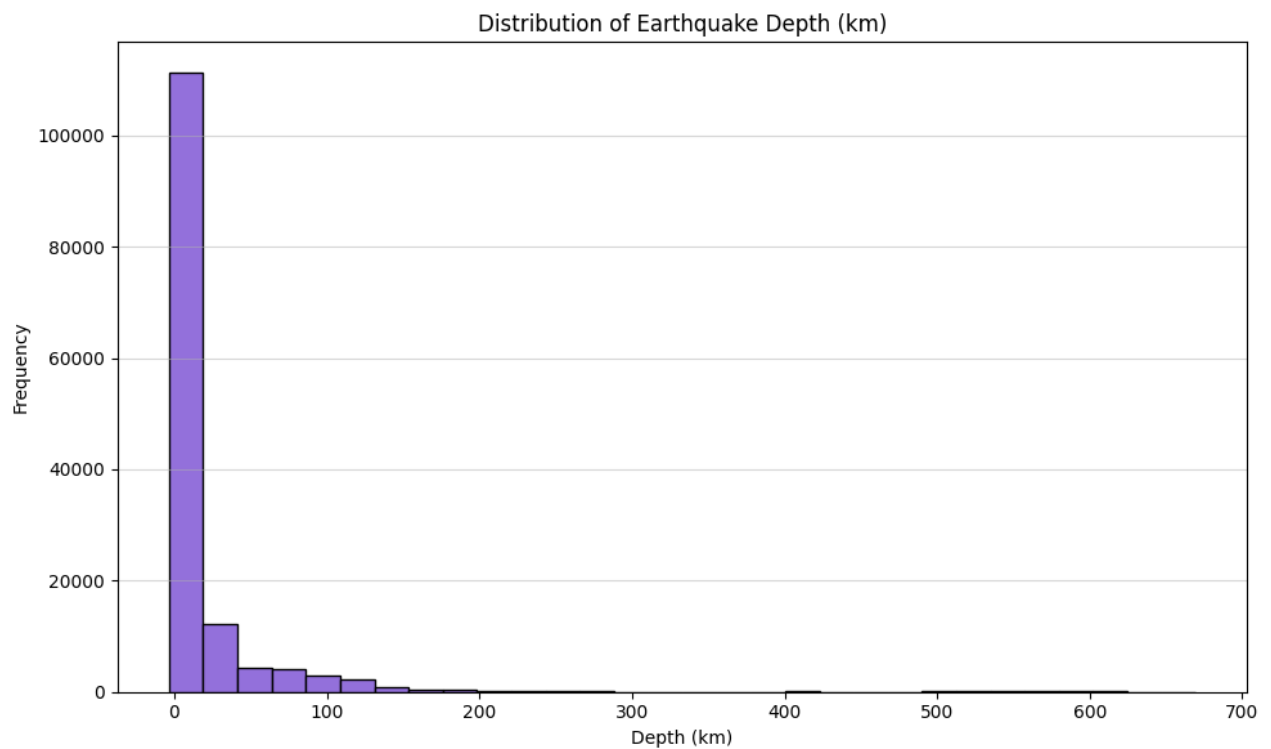
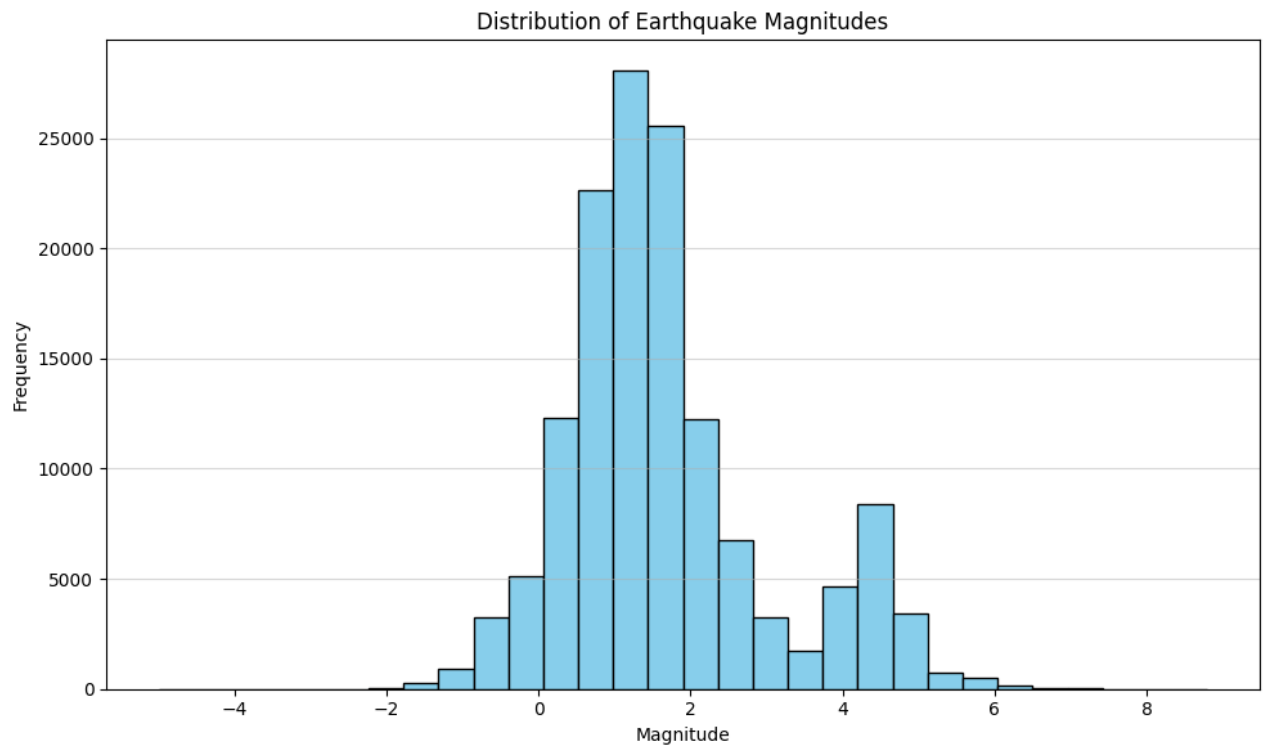
- **Mean Magnitude (~1.67):** The vast majority of recorded activity consists of micro-earthquakes. While often imperceptible to humans, these are critical for defining the density of seismic zones for clustering.
- **Maximum Magnitude (8.8):** The dataset successfully captures rare, high-impact global events.
- **Negative Magnitudes:** The presence of negative values is preserved. These represent extremely small "micro-earthquakes" detected by sensitive local sensors and are valid data points for density-based clustering.
- **Missing Data:** Only ~0.003% of records (5 out of 140k) were missing magnitude data, which is statistically negligible.

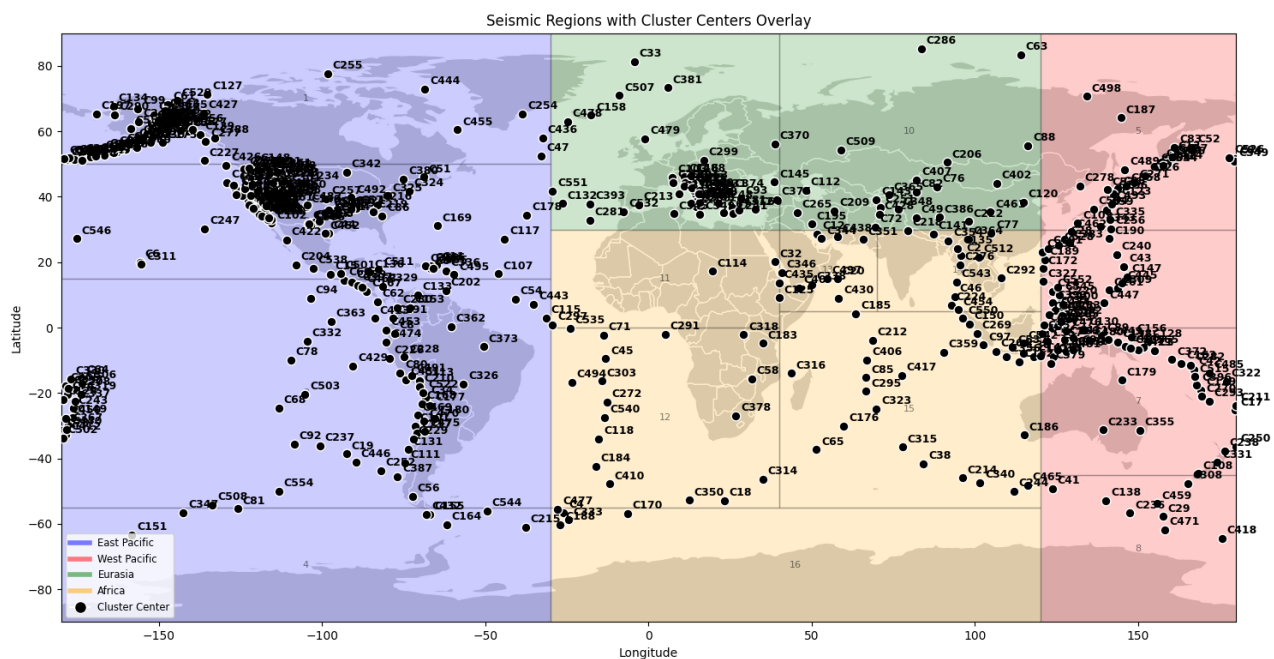
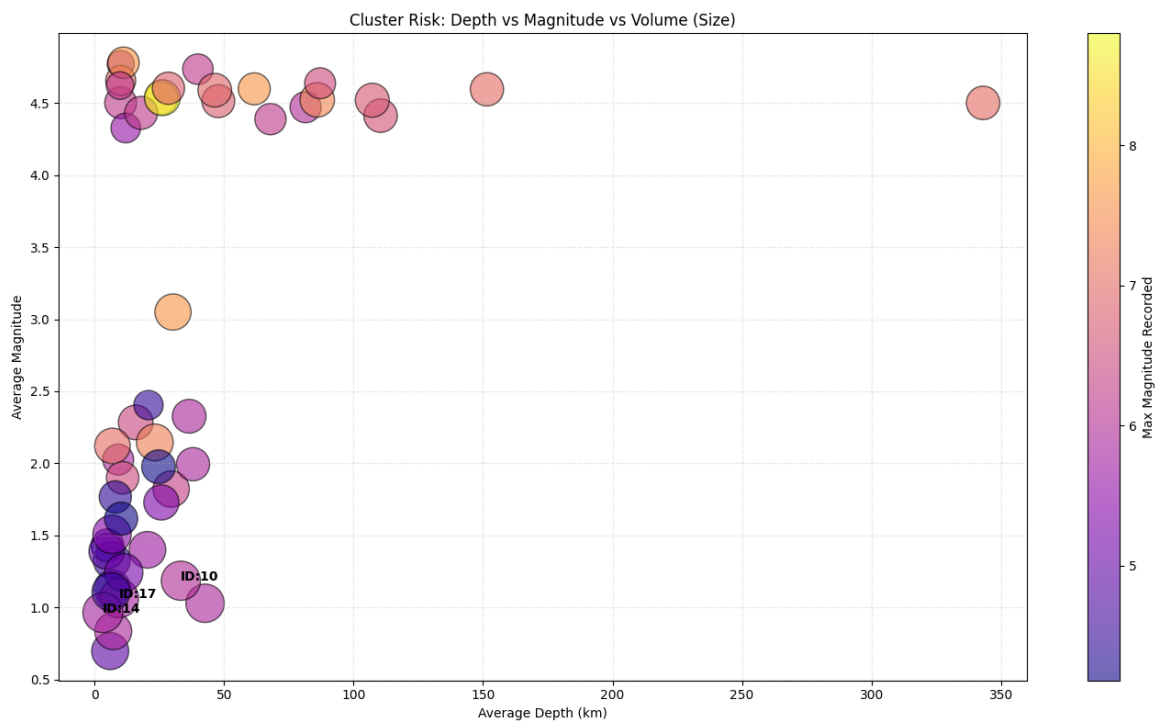


Depth Distribution

The depth analysis confirms the dataset's geological validity, capturing the full range of seismic zones from the surface to the mantle.

- **Shallow vs. Deep:** The distribution is heavily skewed towards shallow crustal events (0–70 km), but the maximum recorded depth of ~670 km confirms the successful capture of deep-focus earthquakes in subduction zones (transition zone/lower mantle boundary).
- **Negative Depths:** A small subset of data shows negative depth values. These were validated as correct readings indicating events occurring above the reference geoid (e.g., seismic activity inside mountains or volcanic structures).





System Architecture

1. Architectural Pattern: Lambda Architecture

The system is built on the Lambda Architecture design pattern. This approach was chosen to solve the specific challenge of earthquake monitoring: the need to process massive volumes of historical data for accurate pattern recognition (Batch Layer) while simultaneously providing low-latency alerts for ongoing events (Speed Layer).

The architecture allows the system to balance throughput, latency, and fault tolerance by decoupling real-time processing from comprehensive historical analysis.

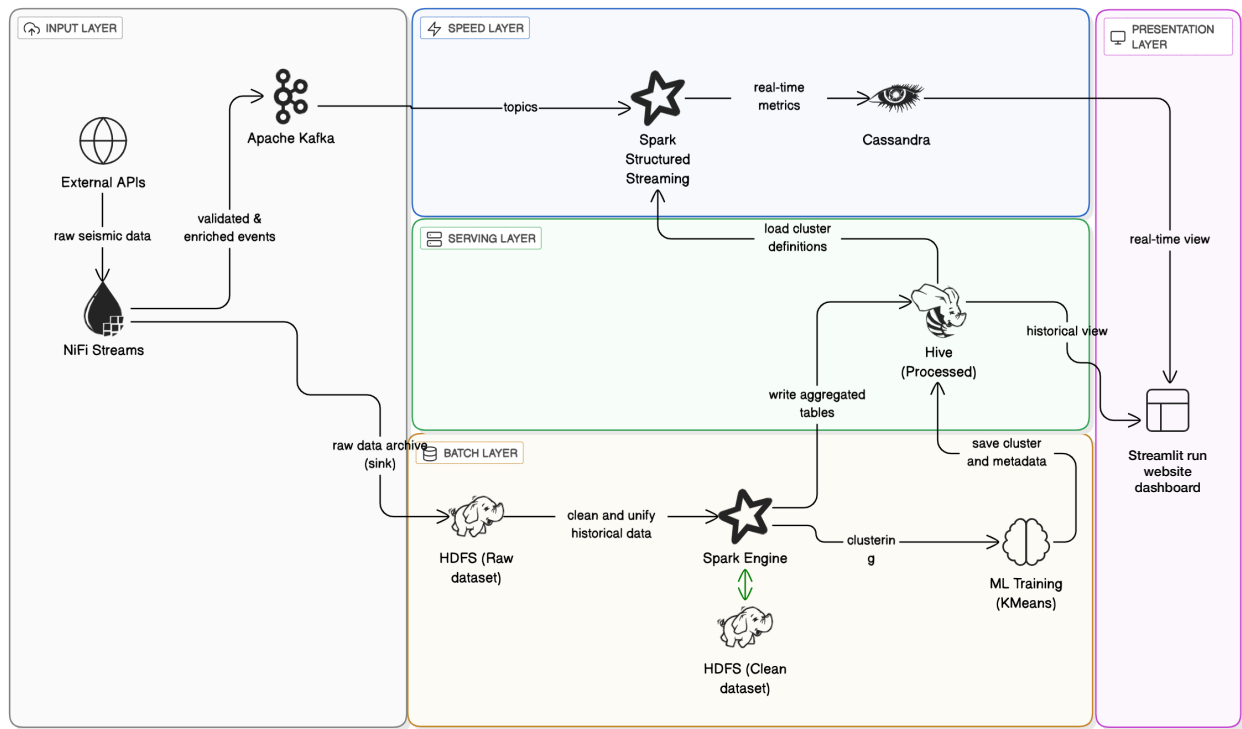
2. Layer-by-Layer Description

The system consists of five distinct layers, each handling a specific stage of the data lifecycle:

1. Input Layer (Ingestion & Routing)

- Technologies: Apache NiFi, Apache Kafka

- Function: Apache NiFi acts as the robust entry point, managing scheduled API calls to USGS and Terraquake. It implements a Forked Routing Strategy:
 - Hot Path: Cleaned, standardized JSON events are immediately pushed to Apache Kafka topics to feed the real-time engine.
 - Cold Path: Raw response files are written directly to HDFS (Raw Layer). This ensures an immutable historical archive is preserved before any transformation logic is applied.
2. Speed Layer (Stream Processing, Apache Cassandra)
- Technologies: Apache Spark Structured Streaming, Apache Cassandra
 - Function: This layer prioritizes low latency over complex aggregation.
 - It consumes the "Hot Path" data from Kafka.
 - It performs lightweight parsing and immediate enrichment (e.g., assigning a region and cluster based on the ML model).
 - It bypasses heavy deduplication or ML training to ensure speed.
 - Storage: The output is written to Apache Cassandra, a NoSQL database chosen for its high-speed write performance, enabling the dashboard to display events seconds after detection.
3. Batch Layer (Processing & ML)
- Technologies: Apache Spark (PySpark), HDFS, Apache Hive
 - Function: This layer manages the Master Dataset and performs the heavy computational lifting.
 - ETL: It reads from the "Cold Path" (HDFS), performing rigorous cleaning, deep deduplication (fuzzy matching), and schema unification to create the Silver Layer.
 - Machine Learning: It retrains the K-Means Clustering model daily on the full historical dataset. This "re-learning" ensures that the definition of seismic risk zones evolves as new tectonic patterns emerge.
 - Aggregation: It generates the daily statistical summaries (Gold Layer) for regions and clusters.
4. Serving Layer
- Technologies: Apache Hive
 - Function: This layer provides a unified interface for the frontend to query data without needing to understand the underlying complexity of the storage files.
 - Hive: Serves complex analytical queries.
5. Presentation Layer
- Technology: Streamlit (Python)
 - Function: The user interface where insights are consumed. It connects to both the Serving and Speed layers to render the interactive dashboard.
 - Real-Time View: Pulls live event data from Cassandra.
 - Analytical View: Pulls long-term trend lines and cluster centres from Hive.



eraser

Data Ingestion and Transformation Modules

1. Data Ingestion Strategy (Apache NiFi)

To handle API constraints and high-throughput requirements, the ingestion layer uses a Forked Routing Strategy:

- Hot Path (Speed): Sends simplified JSON to Apache Kafka for immediate streaming.
- Cold Path (Batch): Writes raw response files to HDFS to preserve an immutable historical archive.

2. Batch Transformation (PySpark)

The Batch Layer creates the "Silver Layer" (Clean Data) through a rigorous cleaning pipeline:

- Schema Unification: Casts diverse JSON sources to a strict schema, standardizing timestamps (BigInt) and coordinates (Double).
- Advanced Deduplication: Merges USGS and Terraquake data using Fuzzy Matching. Events are considered identical if they fall within the same 1-minute time bucket and ~11km geographic bucket.
- Handling Missing Data: Records missing latitude or longitude are dropped, as they are unusable for spatial clustering.
- Aggregation: Generates two Gold Layer views: Fixed Region (based on static boundaries) and Dynamic Cluster (based on ML output).

3. Speed Transformation (Streaming)

The Speed Layer prioritizes latency over deep cleaning using Spark Structured Streaming:

- Lightweight Processing: Performs immediate parsing, basic region assignment and cluster classification on the Kafka stream.
- Idempotent Storage: Writes to Apache Cassandra. Using the Event ID as the Primary Key allows the system to overwrite duplicates automatically, avoiding slow "read-before-write" checks.

Analytical Module

1. Core Task: Unsupervised Seismic Clustering

The Analytical Module employs unsupervised machine learning to partition global seismic events into distinct spatial clusters. This transforms noisy, individual data points into better analytically centres (clusters).

Algorithm: K-Means Clustering (via PySpark MLlib).

Why K-Means? Selected for its scalability ($O(n)$ complexity) and efficiency in handling large-scale distributed datasets compared to density-based algorithms like DBSCAN.

2. Feature Engineering: Spherical Projection

A critical technical challenge was the distortion caused by standard Euclidean distance when applied to a sphere (the Earth). Using raw Latitude/Longitude creates errors near the Poles and the International Date Line (where -180° and $+180^\circ$ are numerically distant but geographically adjacent). To solve this, spherical coordinates are converted into 3D Cartesian vectors (x, y, z) on a unit sphere.

- $x = \cos(\text{lat}) \cdot \cos(\text{lon})$
- $y = \cos(\text{lat}) \cdot \sin(\text{lon})$
- $z = \sin(\text{lat})$

This ensures that cluster centroids accurately reflect physical proximity, eliminating "Date Line" edge cases.

3. Training Strategy & Model Selection

Since K-Means requires the number of clusters (k) to be defined a priori, the system implements an Automated Tuning Pipeline that runs during every daily batch process:

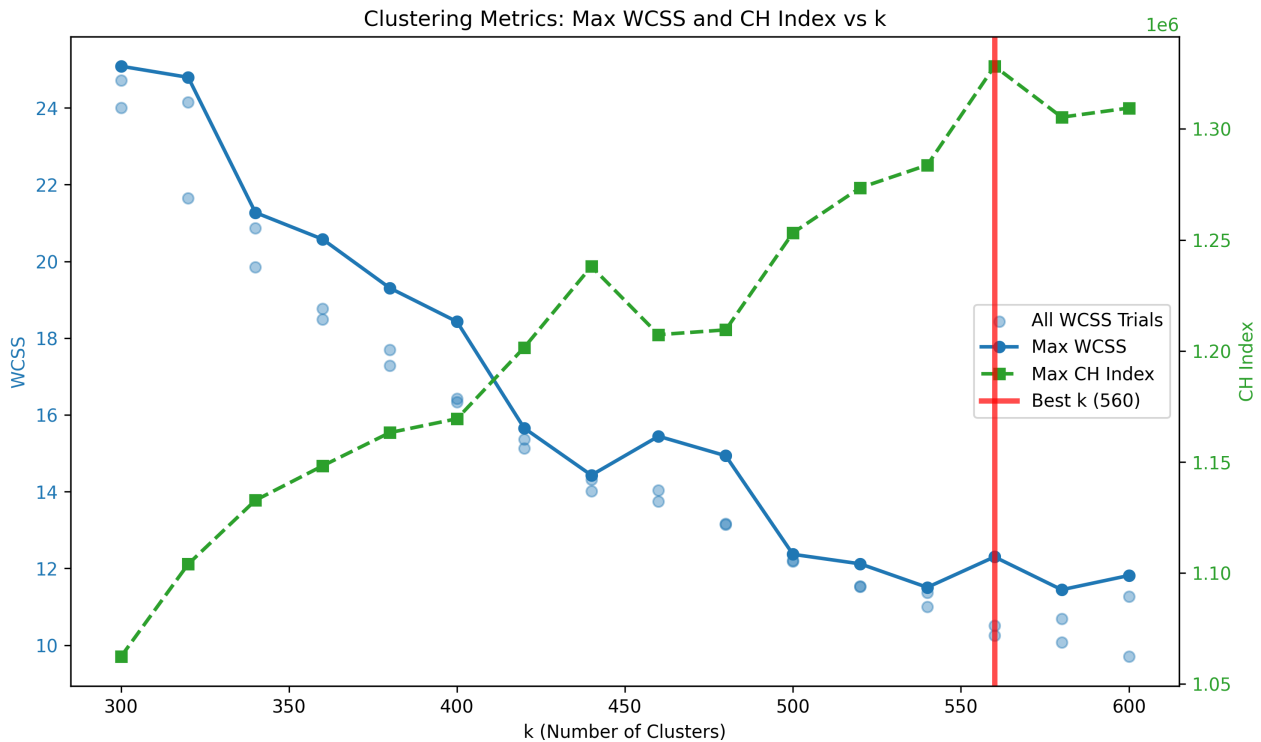
Iterative Grid Search: The model trains across a range of k values (e.g., k=300 to 600) to empirically find the optimal granularity for the current global seismic state.

Stability Trials: To mitigate K-Means' sensitivity to initial random centroids, the algorithm runs 3 independent trials for each k. Only the model with the lowest Within-Cluster Sum of Squares (WCSS) is retained for evaluation.

4. Key Performance Metric: Calinski-Harabasz Index

To automatically select the "Global Best Model" from the grid search, the system uses the Calinski-Harabasz (CH) Index. The CH Index measures the ratio of Inter-Cluster Dispersion (separation between clusters) to Intra-Cluster Dispersion (compactness within a cluster).

A higher CH score indicates better-defined clusters. The CH Index was chosen over the Silhouette Score because it is significantly more computationally efficient ($O(N)$ vs $O(N^2)$), which is critical when retraining on large historical datasets daily.



Project Results and Business Assessment

1. Summary of Project Outcomes

The project has successfully transitioned from a conceptual design to a fully operational Real-Time Earthquake Monitoring System. By implementing a robust Lambda Architecture, the solution delivers on its core promise: converting fragmented, high-volume global seismic data into a unified and actionable intelligence feed.

2. Key Technical Achievements:

- **Operational Lambda Pipeline:** Successfully deployed a dual-layer architecture where Apache NiFi and Kafka manage real-time ingestion, while Spark and HDFS handle complex historical processing.
- **Unified Global Dataset:** Created a high-quality "Silver Layer" that merges and deduplicates over 140,000 events from USGS and Terraquake, solving the challenge of conflicting data providers.
- **ML-Driven Risk Zoning:** Implemented an unsupervised K-Means Clustering engine that dynamically partitions the globe into active "Risk Zones," retraining daily to reflect evolving tectonic realities.
- **Multi-Persona Dashboard:** Delivered a functional Streamlit interface that serves distinct views for immediate situational awareness (Real-Time) and long-term research (Analytical).

3. Business Value Evaluation

From a business perspective, the solution offers significant improvements over traditional, manual monitoring methods:

- **Reduction of Cognitive Load:** By replacing raw event lists with Intelligent Clustering, the system reduces "alert fatigue." Operators are no longer overwhelmed by thousands of micro-tremors; instead, they can focus on significant anomalies within established clusters.
- **Operational Reliability:** The Dual-Source Strategy mitigates the risk of relying on a single government agency. If one API experiences downtime, the secondary source ensures continuous situational awareness.
- **Process Automation:** The automated ETL pipeline eliminates hours of manual data cleaning and spreadsheet management, allowing analysts to focus entirely on high-value interpretation and reporting.

Impact Analysis

1. Societal Impact

The primary societal benefit of this Big Data module is enhanced Public Safety and Preparedness. **Early Warning & Response:** By minimizing the latency between event detection and visualization (via the Speed Layer), the system empowers Emergency Operations Officers to initiate response protocols faster. In disaster scenarios, gaining even minutes can significantly impact evacuation success. The system's ability to visualise historical clusters helps demystify seismic activity. It transforms abstract risk into concrete visual data, potentially aiding urban planners and policymakers in communicating risk to populations living in fault zones.

2. Economic Impact

- **Optimized Resource Allocation:** Disaster response is expensive. By identifying "hotspots" and separating noise from significant trends, governments and NGOs can pre-position resources (medical supplies, search teams) more effectively, reducing the economic waste of reactive mobilization.
- **Risk Modeling:** The "Gold Layer" aggregated data provides a foundation for more accurate risk assessment. This high-fidelity historical data could be valuable for the insurance and construction sectors to refine premiums and building standards in identified high-density clusters.

3. Ethical and Social Considerations

- **Algorithmic Responsibility:** The system uses Machine Learning to define "Risk Zones." There is an ethical responsibility to ensure these clusters are accurate. A "False Positive" (identifying a new high-risk zone erroneously) could cause unnecessary panic, while a "False Negative" could lead to complacency. The implementation of the Stability Trials and Calinski-Harabasz validation metrics mitigates this risk.
- **Global Equity:** Seismic sensors are often denser in wealthy nations. By unifying data from multiple global providers (USGS and Terraquake), the system helps bridge the data gap, ensuring that seismic activity in developing regions is captured and visualized with the same rigor as in heavily monitored areas.
- **Data Integrity:** As a "Single Source of Truth," the system bears the ethical burden of accuracy. The rigorous deduplication logic ensures that the system does not over-report the frequency of earthquakes, which could otherwise inflate risk perceptions.